

AUTONOMOUS MEDICAL SUPPLY DELIVERY ROBOT FOR HOSPITALS



A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Manual transport of medicines and supplies between hospital stores and wards places a heavy burden on nursing staff, slows delivery during busy periods, and raises infection-transmission risk between departments. This project addresses that gap through an Autonomous Medical Supply Delivery Robot capable of navigating hospital corridors and delivering payloads without human guidance. Built on an Arduino UNO, the system combines infrared sensors for path tracking, an ultrasonic sensor for obstacle detection, and a differential-drive motor arrangement for movement, with an onboard compartment for carrying supplies and an LCD for status feedback. Testing on a simulated hospital corridor confirmed accurate path-following, dependable obstacle avoidance, and collision-free payload delivery across repeated trials. The prototype maintained a stable route with minimal drift, halted consistently within a safe distance of obstacles, and carried moderate loads without affecting navigation performance. Battery evaluation further showed steady power delivery over an extended working period, supporting sustained operation between charging cycles. Fabricated at a cost of roughly Rs. 2,820, the prototype proves that embedded, sensor-based automation offers a practical, budget-friendly route to safer, more efficient hospital logistics. Planned enhancements such as wireless monitoring, automated ward recognition, and vision-guided navigation could extend this design toward full-scale hospital deployment.

Keyword(s): Autonomous Mobile Robot, Hospital Logistics, Line Following, Obstacle Avoidance, Arduino UNO, Medical Supply Delivery, Healthcare Automation

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LIST OF SYMBOLS, ABBREVLATIONS AND NOMENCLATURE

1. SYMBOLS

SYMBOL	DESCRIPTION	UNIT
V	Battery voltage	V
I	Current consumed by the robot	A
P	Electrical power	W
N	Motor speed	rpm
PWM	Pulse Width Modulation duty cycle	%
D	Distance travelled by the robot	m
d	Distance between robot and obstacle	cm
v	Robot travelling speed	m/s
t	Operating time	s
T	Motor torque	N·m
W	Payload weight	kg
C	Battery capacity	Ah
V_m	Motor voltage	V
I_m	Motor current	A
S_l	Left IR sensor status	—
S_r	Right IR sensor status	—
LCD	Liquid Crystal Display	—
UNO	Arduino UNO controller	—
RPM	Revolutions per minute	rpm
θ	Turning angle of the robot	degree (°)

2. ABBREVIATIONS

ABBREVIATION	FULL FORM
AMR	Autonomous Mobile Robot
IR	Infrared
PWM	Pulse Width Modulation
LCD	Liquid Crystal Display
DC	Direct Current
I2C	Inter-Integrated Circuit
SLAM	Simultaneous Localization and Mapping
RFID	Radio Frequency Identification
IoT	Internet of Things
HC-SR04	Ultrasonic Distance Sensor Module (model designation)
IDE	Integrated Development Environment
PPE	Personal Protective Equipment

3. NOMENCLATURE

TERM	DEFINITION / DESCRIPTION
Autonomous Robot	Robot that performs movement and delivery tasks with minimal human intervention
Medical Supply	Medicines and essential healthcare materials transported by the robot
Obstacle Detection	Process of identifying objects in the robot's path
Ultrasonic Sensor	Sensor used to measure the distance between the robot and an obstacle
IR Sensor	Infrared sensor used to detect the path or line for robot navigation
Line Following	Method of guiding the robot by detecting a predefined path
Arduino UNO	Microcontroller used to control the robot's sensors and motors
Motor Driver	Electronic module used to control the speed and direction of DC motors
DC Gear Motor	Motor used to provide movement and driving force to the robot
LCD Display	Display used to indicate the operating status of the robot
Push Button	Input device used to start or stop the robot
Navigation	Process of guiding the robot from one location to another
Obstacle Avoidance	Process of stopping or changing the robot's movement when an obstacle is detected
Payload	Medical supplies carried by the robot
Battery Pack	Power source used to operate the robot
Voltage Regulator	Device used to provide a suitable voltage to electronic components
Prototype Model	Small-scale working model developed to demonstrate the proposed robot system

CHAPTER 1

INTRODUCTION

Modern hospitals handle a very high volume of internal logistics every day, including the movement of medicines, injections, dressing materials, laboratory samples, and other small consumables between the central pharmacy or store and individual wards. This task is presently carried out almost entirely by nurses and support staff walking between departments, which consumes valuable clinical time that could otherwise be spent on direct patient care. During periods of high patient load, infectious disease outbreaks, or staff shortages, manual transport of supplies becomes a significant bottleneck and also increases the risk of person-to-person transmission of infection between wards.

Autonomous Mobile Robots (AMRs) have emerged as a practical solution to this problem in hospitals across the world, being used to transport medicines, food trays, laboratory samples, and linen between departments with minimal human contact. Existing commercial hospital delivery robots, however, are typically expensive, proprietary systems that are not easily accessible for small and medium-sized hospitals, particularly in developing regions. This creates a strong motivation to explore low-cost, embedded-systems-based alternatives that can achieve reliable autonomous navigation within a hospital-like environment using inexpensive and widely available components.

This project presents the design and development of an Autonomous Medical Supply Delivery Robot intended to transport medical supplies along a fixed corridor route between a supply point and a ward without requiring an operator to drive or guide it. The prototype uses an infrared (IR) sensor array to continuously track a visually marked path on the floor, an Arduino UNO microcontroller to process sensor data and control motor movement, and a differential-drive wheel arrangement to steer the robot along the path. A 16x2 LCD display provides real-time operating status, while an HC-05 Bluetooth module allows a staff member to monitor the robot's status from a nearby smartphone or laptop.

A scaled prototype was fabricated and tested on a simulated hospital corridor, created using adhesive tape markings on a white sheet to represent a typical straight-and-turn ward corridor layout. The prototype's navigation accuracy, path-tracking stability, and power consumption were evaluated through repeated trial runs. The primary objective of this project is to demonstrate that a low-cost, sensor-guided autonomous robot can reliably perform point-to-point delivery of medical supplies within a hospital corridor, thereby reducing staff workload, minimizing unnecessary physical contact between departments, and improving the speed and consistency of hospital material handling.

1.1 PROBLEM STATEMENT

Hospitals require the continuous movement of medicines, consumables, and small medical supplies between the central pharmacy or store and individual wards. This task is presently performed manually by nurses and support staff, which consumes clinical time that could otherwise be spent on direct patient care. Manual transport also introduces variability in delivery time, increases staff workload during peak hours, and raises the risk of person-to-person transmission of infection when the same staff move between multiple wards. There is a clear need for a low-cost, reliable system that can carry out routine point-to-point supply delivery within a hospital corridor without requiring a human operator to steer it and without colliding with people or equipment along the way.

1.2 OBJECTIVES

- To design and fabricate a low-cost autonomous mobile robot capable of navigating a fixed hospital corridor route.
- To implement reliable line-following navigation using infrared (IR) sensors.
- To integrate an ultrasonic sensor for real-time obstacle detection and safe stopping.
- To provide a dedicated compartment for safely carrying medical supplies during transit.
- To provide local status indication through an LCD display and simple start/stop control through a push button.
- To experimentally evaluate the navigation accuracy, obstacle response, payload capacity, and battery performance of the prototype.

1.3 SCOPE OF THE PROJECT

The scope of this project covers the design, fabrication, and experimental evaluation of a scaled prototype of an autonomous medical supply delivery robot. The prototype is developed and tested on a simulated hospital corridor track built using tape markings on a flat surface, with obstacles deliberately introduced to evaluate the obstacle detection system. The project covers embedded hardware selection, circuit design, control algorithm development, and trial-based performance testing. It does not cover full clinical deployment, integration with hospital information systems, or regulatory certification, which are identified as directions for future work.

1.4 NEED FOR THE PROPOSED SYSTEM

Commercial hospital delivery robots such as Panasonic's HOSPI and Aethon's TUG have demonstrated the practical benefits of autonomous delivery in real hospitals, including reduced staff workload and reduced physical contact between departments. However, these systems are

proprietary and expensive, placing them out of reach for small and medium-sized hospitals, particularly in developing regions. There is a clear need for an affordable, embedded-systems-based alternative that can achieve dependable navigation and basic obstacle avoidance using inexpensive, widely available components, while still meeting the core requirement of safe, contact-free supply transport within a hospital corridor.

1.5 METHODOLOGY

This project adopts a systematic methodology to design, fabricate, and evaluate the proposed autonomous delivery robot, integrating literature review, hardware and circuit design, embedded programming, and experimental trial runs.

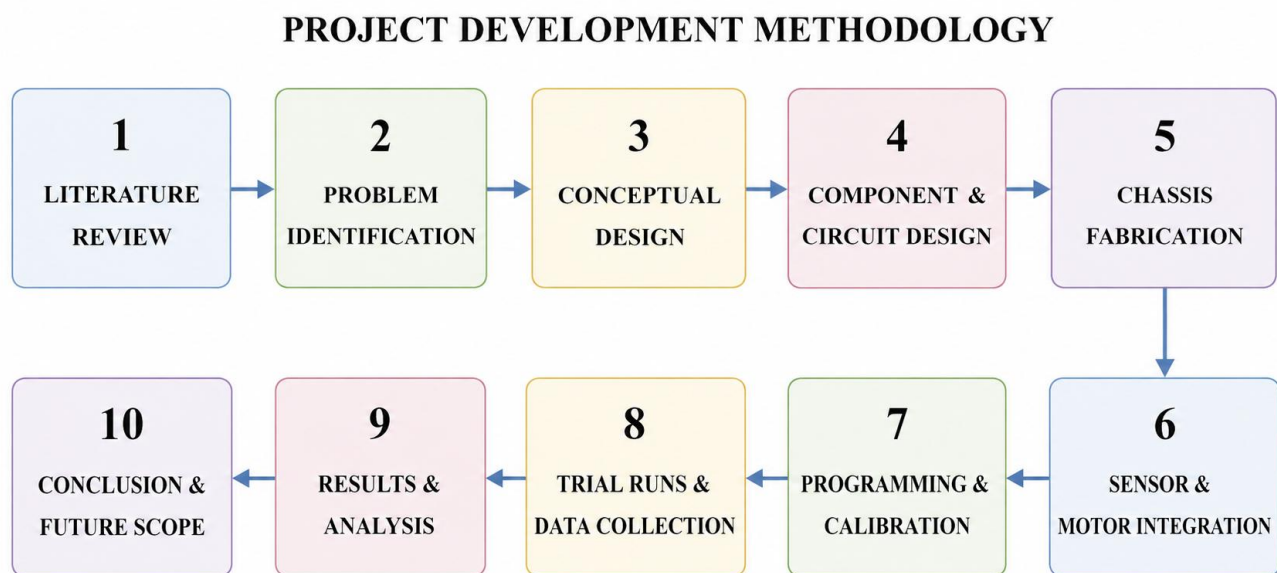


Figure 1.1 Methodology of the proposed system

The process began with a review of existing hospital delivery systems and navigation techniques, followed by problem identification and definition of project objectives. A conceptual design was then developed, components were selected, and the chassis was fabricated with sensor and motor mounting positions. Control firmware combining line-following and obstacle-avoidance logic was developed and calibrated, after which the prototype was tested through repeated trial runs on the simulated corridor track, with the resulting data analyzed to draw conclusions and identify future improvements.

CHAPTER 2

LITERATURE REVIEW

- **Development of a Mobile Robot for Distribution of Medicine in Hospitals (2024)**

The authors developed a mobile robot specifically for medicine distribution in hospital environments. The study focused on the mechanical design and working principle of the robot and demonstrated a medicine-delivery scenario using path planning techniques. The research highlighted the potential of mobile robots to improve hospital logistics and automate the transportation of medicines to patient rooms. The study also identified path planning as an important factor for reliable hospital delivery operations.

- **Robotics and Artificial Intelligence in Healthcare During COVID-19 Pandemic (2021)**

The researchers reviewed the applications of robotics and artificial intelligence in healthcare during the COVID-19 pandemic. The study reported that autonomous robots can support hospital supply and delivery operations by transporting medicines, medical samples, food, and other materials. The research also highlighted that autonomous delivery robots can reduce the workload of healthcare workers and support contactless delivery, particularly in situations where minimizing human interaction is important.

- **Stochastic Scheduling of Autonomous Mobile Robots at Hospitals (2023)**

The authors investigated the scheduling of autonomous mobile robots used for hospital medicine delivery. The study considered the transportation of medicines from pharmacies or warehouses to different hospital wards and addressed uncertainties in travel and service times. An improved Tabu Search algorithm was proposed to optimize robot scheduling and reduce operational costs and late deliveries. The research demonstrated the importance of efficient task scheduling when autonomous robots are used for hospital logistics.

- **Boosting the Hospital by Integrating Mobile Robotic Assistance Systems (2024)**

The authors studied the integration of mobile robotic assistance systems into hospital workflows and analysed the risks associated with their deployment. The research emphasized that hospital environments have unique requirements involving people, equipment, corridors, and operational procedures. The study concluded that understanding and managing these risks is essential for safe and reliable operation of mobile robots in hospitals.

- **Autonomous Fever Detection, Medicine Delivery, and Environmental Disinfection for Pandemic Prevention (2023)**

Su and Young developed an autonomous mobile robot system capable of performing multiple healthcare-related tasks, including medicine delivery. The robot was designed for use in hospital fever stations and isolation wards, where contactless operation can reduce direct interaction between healthcare workers and patients. The study demonstrated the potential of combining mobility with autonomous healthcare functions to support hospital operations.

- **A Systematic Review on Autonomous and Cognitive Medical Robots for Acute Care Medicine (2026)**

The researchers reviewed studies on autonomous and cognitive robots used in healthcare applications. The review covered applications including telepresence, therapy, rehabilitation, and medication delivery. The findings indicated that autonomous robots can provide effective support in healthcare environments, while cost, system complexity, and implementation challenges remain important considerations for wider deployment.

- **Autonomous Robot for Medication Delivery in Hospitals Using YOLOv5 Object Detection (2025)**

The researchers developed an autonomous robot for delivering medicine boxes in hospital environments using visual recognition and navigation. The system used YOLOv5 to identify numerical room labels and used ROS2, a camera, and mecanum wheels for indoor navigation. The study demonstrated how vision-based identification can be used to improve autonomous medicine delivery and enable navigation through hospital corridors.

- **Application Management and Effectiveness Analysis of Intelligent Logistics Robots in Hospital Drug and Specimen Delivery Scenarios (2026)**

The authors evaluated an intelligent logistics robot system for drug and specimen delivery in a tertiary hospital. The system used autonomous mobile robots with hierarchical navigation and coordination algorithms. Compared with manual delivery, the robot system reduced delivery time by approximately 32–36% and achieved high verification and item-integrity performance. The study demonstrated the practical benefits of autonomous robots for hospital logistics and medical supply transportation.

- **Feasibility of Autonomous Medication Delivery Robots in High-Traffic Hospital Environments (2026)**

The researchers investigated the feasibility of autonomous medication delivery robots in a real hospital environment, particularly under elevator congestion. The study analyzed 122 non-urgent delivery missions and found that elevator congestion could increase delivery failures and delay. The results highlighted that hospital infrastructure and traffic conditions must be considered when designing and deploying autonomous medical delivery robots.

CHAPTER 3

COMPONENTS SPECIFICATION AND DESCRIPTION

3.1 COMPONENTS

The components used in the fabrication of the scaled prototype are listed in Table 3.1. Each component is described in detail in the following sections.

Table 3.1 Components Used in the Prototype

S.NO	COMPONENT	QUANTITY
1	Arduino UNO (ATmega328P)	1
2	IR sensor module	2
3	Ultrasonic sensor (HC-SR04)	1
4	L298N dual H-bridge motor driver	1
5	DC gear motor 12V with wheels	2
6	Li-ion battery pack (7.4V)	1
7	Voltage regulator (7805)	1
8	16x2 LCD display	1
9	Push button	1
10	Cardboard / acrylic chassis with storage compartment	1 set
11	Jumper wires and connectors	1 set

3.2 COMPONENTS DESCRIPTION

3.2.1 Arduino UNO

The Arduino UNO, based on the ATmega328P microcontroller, serves as the central processing unit of the robot. It reads the analog and digital signals from the IR sensor array and the ultrasonic sensor, executes the navigation and obstacle-avoidance control logic, and drives the motor driver module and LCD display accordingly. The Arduino UNO was chosen for its simple, well-documented architecture, sufficient digital and analog I/O pins for this application, and ease of programming through the Arduino IDE, making it well suited for rapid prototyping of embedded control systems.



Figure 3.2.1 Arduino UNO

Table 3.2 Pin configuration of Arduino UNO

Pin	Function
D2, D3	Digital inputs from left and right IR sensors
D4, D5	Ultrasonic sensor TRIG and ECHO
D6, D7	PWM speed control (ENA, ENB) to motor driver
D8 - D11	Motor direction control (IN1 - IN4)
D12	Push button input (INPUT_PULLUP)
A0 - A5	LCD data and control lines

3.2.2 IR Sensors

Two Infrared (IR) sensor modules are mounted on the underside of the chassis, at the front, to continuously track the black tape path marked on the test corridor. Each sensor emits infrared light from an onboard LED and measures the intensity of light reflected back to a photodiode; a dark surface reflects less infrared light than a light surface, producing a distinct digital output. The difference between the left and right sensor readings is used by the Arduino UNO to determine whether the robot is centered on the path or has drifted to one side, enabling continuous corrective steering.

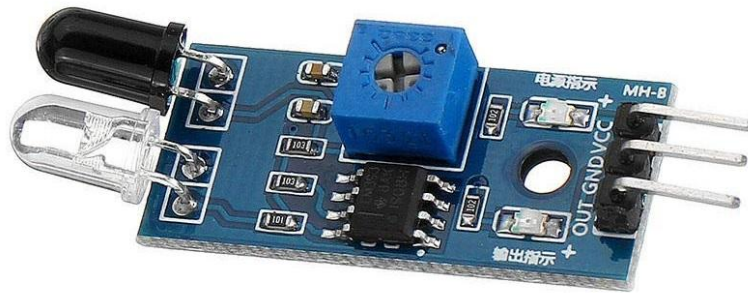


Figure 3.2.2 IR Sensor

Table 3.3 Pin configuration of the IR sensor module

Pin	Function
VCC	Power supply (5V)
GND	Ground
OUT	Digital output (line detected / not detected)

3.2.3 Ultrasonic Sensor

In addition to the two line-tracking IR sensors, an HC-SR04 ultrasonic sensor is mounted at the front of the chassis to detect obstacles in the robot's path. The sensor emits an ultrasonic pulse from its transmitter and measures the time taken for the reflected pulse to return to its receiver; this time-of-flight is converted into a distance measurement, which the Arduino UNO compares against a preset safety threshold to decide whether to stop the robot.

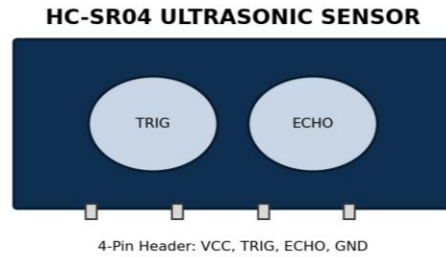


Figure 3.2.3 Ultrasonic Sensor

Table 3.4 Pin configuration of the ultrasonic sensor

Pin	Function
VCC	Power supply (5V)
TRIG	Trigger pulse output from Arduino
ECHO	Echo pulse input to Arduino
GND	Ground

3.2.4 Motor Driver

The L298N is a dual H-bridge motor driver module that allows the Arduino UNO to control the speed and direction of both DC gear motors independently using low-current digital and PWM signals. The module accepts a higher-voltage supply directly from the battery pack to drive the motors, while isolating this higher current path from the microcontroller's logic pins, protecting the Arduino UNO from damage. Independent left-right motor control through this driver is essential for the differential-drive steering used for line-following and obstacle response.

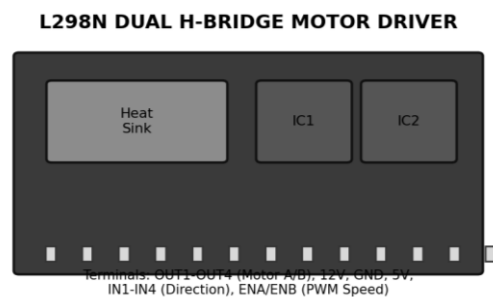


Figure 3.2.4 L298N Motor Driver

3.2.5 DC Motors and Wheels

Two 12V DC gear motors, one for each wheel, provide the drive force for the robot. Each motor converts electrical energy into rotational mechanical motion, and the integrated gearbox reduces output speed while increasing torque, allowing the robot to move at a controlled, steady pace while carrying its payload. Independent speed control of the left and right motors using Pulse Width Modulation (PWM) enables the differential-drive steering that keeps the robot centred on the marked path and allows it to stop promptly when an obstacle is detected.



Figure 3.2.5 DC Gear Motor 12V

3.2.6 Battery and Voltage Regulator

A rechargeable lithium-ion (Li-ion) battery pack supplies power to both the drive motors and the control electronics. Li-ion cells were selected for their high energy density, light weight, and long cycle life, all important for a mobile robot that must operate untethered while carrying a payload. Since the Arduino UNO and sensor modules require a stable 5V supply while the battery voltage varies with charge state, a 7805 linear voltage regulator steps down and stabilizes the supply, protecting the control electronics from voltage fluctuations as the battery discharges.

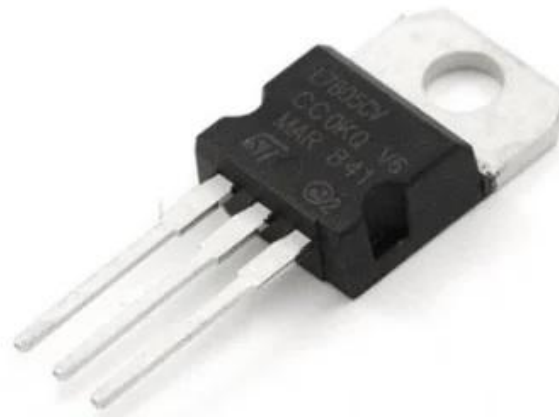


Figure 3.2.6 Li-ion Battery and Voltage Regulator

3.2.7 LCD and Push Buttons

A 16x2 character LCD display is mounted on top of the chassis to show the robot's real-time operating status, such as "System Ready", "Delivering", "Obstacle! Wait", or "Delivery Complete", giving hospital staff a simple visual confirmation of the robot's state without needing a connected device. A momentary push button allows a staff member to start or stop a delivery run directly at the robot, without requiring a separate remote control. When pressed, the button signals the Arduino UNO to toggle the robot between an idle "waiting for dispatch" state and an active "delivering" state.

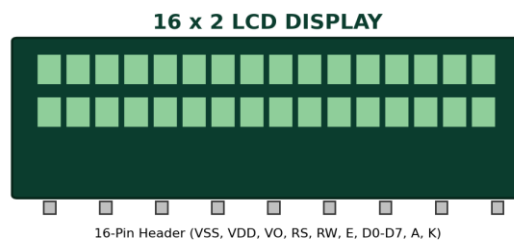


Figure 3.2.7 16x2 LCD Display and Push Button

3.2.8 Medical Storage Compartment

An open compartment is provided on top of the chassis to carry the medical supply payload during transit, sized to accommodate small items such as medicine trays, dressing kits, or sample containers. In the current prototype the compartment is a simple open-top enclosure suitable for validating payload-carrying performance; for real hospital deployment, this compartment would be upgraded to a lockable, subitizable enclosure to secure medicines and prevent tampering or contamination during transit, as discussed further in Chapter 9.

CHAPTER 4

SYSTEM DESIGN AND WORKING PRINCIPLE

4.1 SYSTEM ARCHITECTURE

Figure 4.1 shows the overall system architecture of the robot. The Li-ion battery pack supplies power through the voltage regulator to the Arduino UNO and low-current peripherals, while the motor driver draws power directly from the battery to drive the two DC gear motors. The Arduino UNO reads the IR sensor array, the ultrasonic sensor, and the push button, and controls the motor driver and LCD display, forming a closed control loop between sensing, decision-making, and actuation.

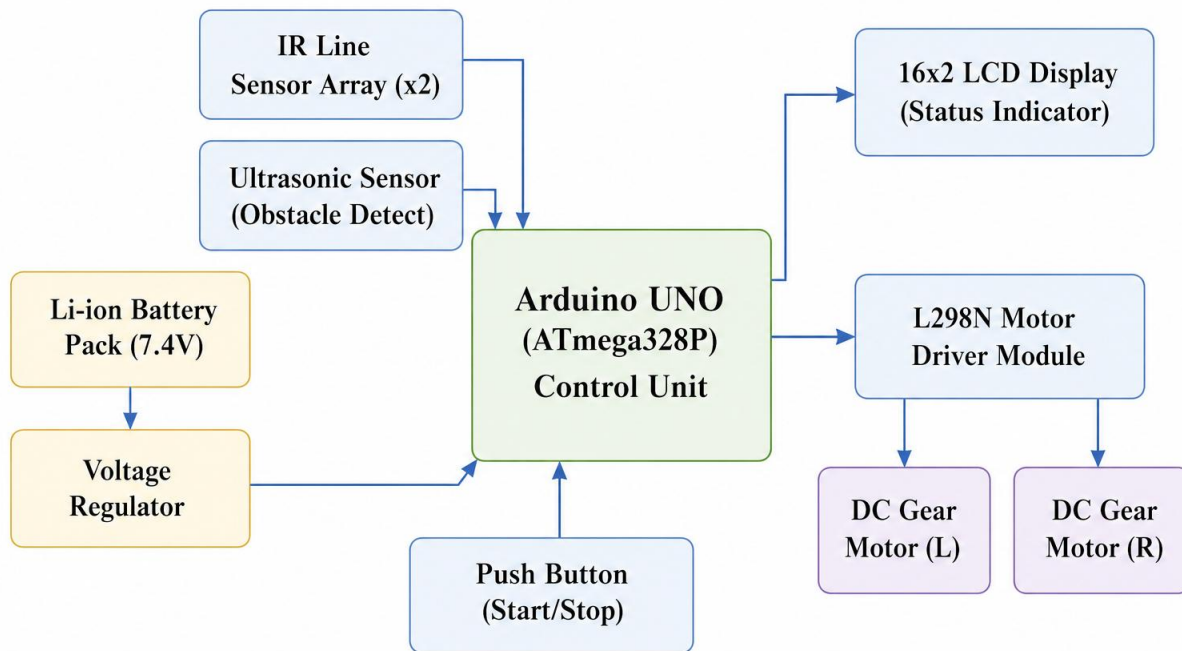


Figure 4.1 System Architecture

4.2 MECHANICAL DESIGN

The chassis is a lightweight two-tier structure: the lower tier houses the drive motors, wheels, and battery pack to keep the center of gravity low and stable, while the upper tier holds the Arduino UNO, sensor breakout boards, LCD display, and the medical storage compartment. The front-mounted IR sensor pair sits close to the ground for reliable line detection, and the ultrasonic sensor is mounted at the front edge of the chassis, facing forward, to detect obstacles before they enter the robot's direct path.

4.3 ELECTRICAL SYSTEM

The electrical system connects all sensors and actuators to the Arduino UNO through direct digital and PWM signal lines, with a separate power path for the motors routed through the L298N driver to avoid loading the microcontroller's regulated supply. Table wiring and pin assignments are summarized in the circuit diagram provided in Appendix 1, which maps each component to its corresponding Arduino UNO pin.

4.4 POWER SUPPLY

Power is supplied by a single 7.4V Li-ion battery pack. This supply is routed in two parallel paths: directly to the L298N motor driver, which regulates the higher-current motor supply internally, and through the 7805-voltage regulator to produce a stable 5V rail for the Arduino UNO, IR sensors, ultrasonic sensor, LCD display, and push button. This dual-path arrangement ensures that motor current surges do not disturb the logic-level supply to the control electronics.

4.5 SENSOR AND MOTOR INTERFACING

The two IR sensors and the ultrasonic sensor connect to dedicated digital pins on the Arduino UNO, as summarized in Chapter 3. The L298N motor driver receives four direction-control signals (IN1-IN4) and two PWM speed-control signals (ENA, ENB) from the Arduino UNO, and in turn drives the two DC gear motors. This interfacing arrangement allows the firmware to independently set the speed and direction of each wheel in response to sensor readings, forming the basis of the differential-drive steering used for navigation.

4.6 NAVIGATION AND OBSTACLE AVOIDANCE

Navigation is achieved by continuously comparing the readings of the left and right IR sensors. When both sensors detect the background surface, the robot is centered on the line and both motors are driven forward at equal speed. When one sensor detects the dark line while the other does not, the robot has drifted off-center, and the control algorithm reduces the speed of the motor on that side while increasing the speed of the opposite motor, steering the robot back onto the path.

Before applying this steering correction on each control cycle, the Arduino UNO first checks the ultrasonic sensor reading. If an object is detected within a preset safety distance (15 cm in this prototype), both motors are stopped immediately regardless of the line-following state, and the LCD display shows an "Obstacle! Wait" message until the obstacle is no longer detected, at which point normal line-following resumes.

4.7 CONTROL ALGORITHM

The line-following behaviour can be expressed using a line-position error term:

$$e = SL - SR$$

where SL and SR are the digital readings of the left and right IR sensors. The corrected motor speeds are then computed as:

$$VL = Vbase - Kp \cdot e \quad VR = Vbase + Kp \cdot e$$

where VL and VR are the PWM speed commands to the left and right motors, Vbase is the nominal forward speed, and Kp is a proportional gain determined experimentally during calibration. This correction is applied only when the ultrasonic sensor confirms the path ahead is clear; otherwise, the obstacle-stop condition overrides the line-following output and both motors are set to zero.

4.8 FLOWCHART

Figure 4.8 presents the complete control flowchart of the robot, combining the push-button-triggered start condition, ultrasonic obstacle check, and IR-sensor-based line-following correction into a single control loop executed continuously while the robot is in the delivering state.

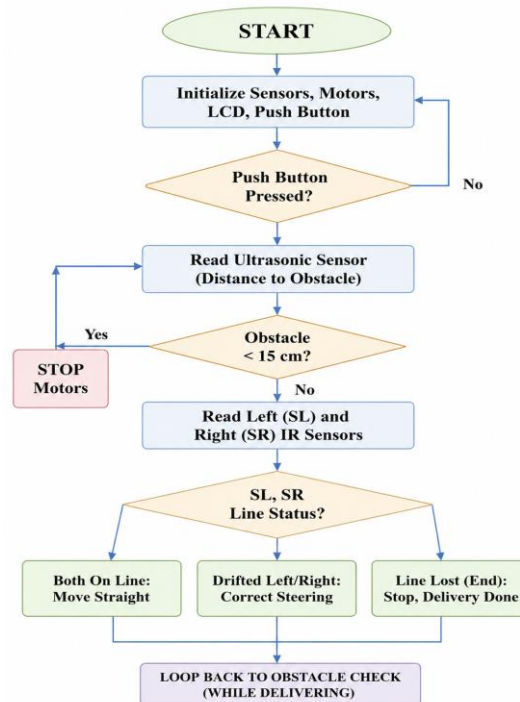


Figure 4.8 Control Algorithm Flowchart

CHAPTER 5

COST ESTIMATION

This chapter presents an itemized cost estimation for the scaled prototype developed in this project, covering component procurement and fabrication. The figures represent indicative market prices at the time of this study.

5.1 COMPONENT COST

Table 5.1 Component Cost

S.NO	COMPONENT	QUANTITY	PRICE (Rs.)
1	Arduino UNO	1	550
2	IR sensor module	2	180
3	Ultrasonic sensor (HC-SR04)	1	120
4	L298N motor driver module	1	150
5	DC gear motor 12V (with wheels)	2	400
6	Li-ion battery pack (7.4V)	1	450
7	Voltage regulator (7805)	1	40
8	16x2 LCD display	1	180
9	Push button	1	20
10	Jumper wires and connectors	1 set	80
Component Sub-Total			2170

5.2 FABRICATION COST

Table 5.2 Fabrication Cost

S.NO	ITEM	PRICE (Rs.)
1	Cardboard and acrylic sheet for chassis	300
2	Adhesive, glue gun and glue sticks	150
3	Fasteners, mounts and miscellaneous hardware	120
4	Tape and materials for test track	80
5	Fabrication Sub-Total	650

5.3 TOTAL PROJECT COST

Table 5.3 Total Project Cost

S.NO	CATEGORY	PRICE (Rs.)
1	Component Sub-Total	2170
2	Fabrication Sub-Total	650
3	Total Project Cost	2820

The total cost of approximately Rs. 2,820 for the scaled prototype confirms that the proposed system can be built at a fraction of the cost of commercial hospital delivery robots, supporting the project's goal of demonstrating an affordable autonomous delivery solution.

CHAPTER 6

MEDIROUTE HOSPITAL DISPATCH WEBSITE

6.1 WEBSITE OVERVIEW

The Medi route Hospital Dispatch Website is designed to provide a simple and organized interface for managing medical supply delivery within the hospital. The website allows authorized hospital staff to select the required medical supply and destination department. The dispatch information can then be used to initiate the delivery process through the autonomous medical supply delivery robot. The system is intended to reduce manual effort and improve the efficiency of routine medical supply transportation between hospital departments. The website provides a centralized interface for requesting, dispatching, and monitoring medical supplies.

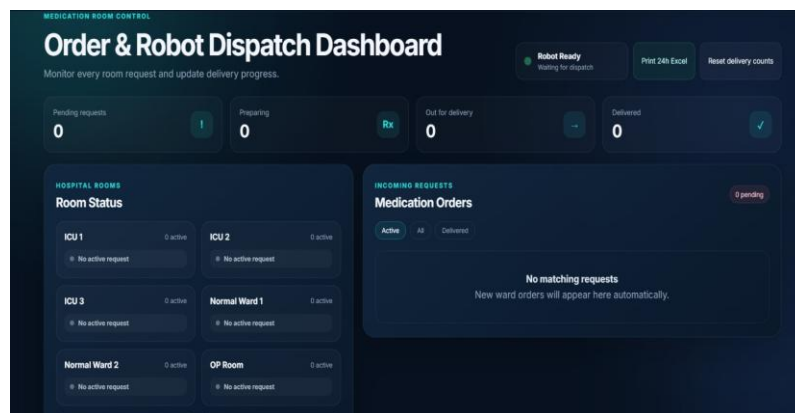


Figure 6.1 Website Overview

6.2 USER LOGIN AND AUTHENTICATION

The login module provides secure access to the Medi Route website. Authorized hospital staff can enter their username and password to access the system. Authentication helps prevent unauthorized users from creating or modifying medical supply requests. After successful login, the user is directed to the main dashboard, where the available dispatch options and hospital departments can be accessed.

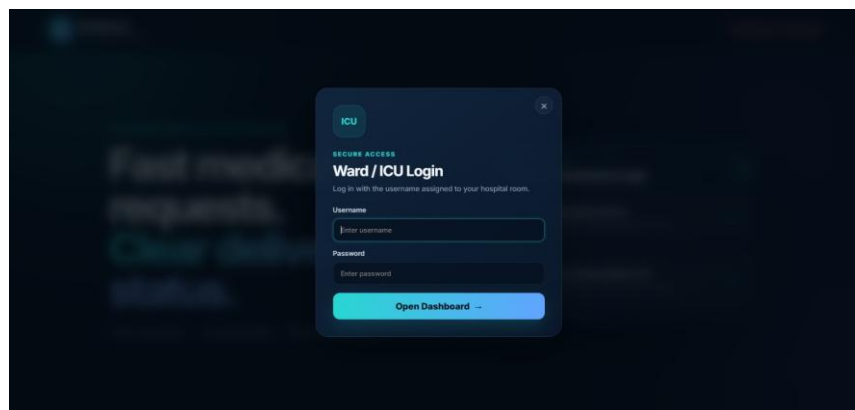


Figure 6.2 User Login and Authentication

6.3 MEDICATION ROOM / DISPATCH MODULE

The Medication Room or Dispatch Module is used to create a medical supply delivery request. The user can select the required medicine or medical supply and provide the necessary details. After selecting the supply, the user can choose the destination department and confirm the dispatch request. This information forms the basic input for the robot delivery process.

6.4 ICU AND WARD SELECTION

The website provides destination selection for different hospital areas such as ICU, general wards, and nursing stations. The user selects the required destination before confirming the delivery request. This feature helps the delivery system identify where the medical supply needs to be transported and supports organized movement of the robot within the hospital.

Figure 6.4 ICU and Ward Selection

6.5 OP, EMERGENCY AND OPERATION THEATRE MODULES

The website can provide separate options for important hospital departments such as Out-Patient (OP), Emergency, and Operation Theatre (OT). These modules help staff specify the destination according to the medical requirement. Emergency-related requests can be identified separately so that urgent medical supplies can be handled with appropriate priority in the proposed system.

6.6 MEDICAL SUPPLY DISPATCH WORKFLOW

The overall dispatch process follows a simple sequence:

User Login → Select Medical Supply → Select Destination → Confirm Request → Robot Dispatch → Delivery → Status Update

The user first logs into the website and selects the required medical supply. The destination department is then selected and the request is confirmed. The robot receives or follows the corresponding delivery instruction and moves towards the selected destination. After completing the delivery, the delivery status can be updated.

6.7 WEBSITE–ROBOT INTEGRATION

The website acts as the user interface for initiating the medical supply delivery process, while the robot performs the physical transportation. The dispatch information selected through the website can be associated with the robot's control system. The robot uses its sensors for navigation and obstacle detection while travelling through the hospital environment. This integration provides a link between the software-based dispatch system and the physical autonomous robot.

6.8 SOFTWARE SYSTEM ARCHITECTURE

The software architecture consists of the user interface, dispatch module, control system, and autonomous robot. The user interacts with the website to create a delivery request. The request is processed by the control system and associated with the robot operation. The basic architecture can be represented as:

Hospital Staff → MediRoute Website → Dispatch/Control System → Robot Controller → Sensors & Motors → Medical Supply Delivery

This architecture provides a structured approach for connecting hospital supply requests with autonomous robot operation.

6.9 DISPATCH STATUS AND MONITORING

The dispatch status module can be used to indicate the current stage of a delivery request. Typical status stages include Requested, Dispatched, In Transit, and Delivered. Status monitoring helps hospital staff understand whether the requested medical supply has been dispatched and whether the delivery has been completed.

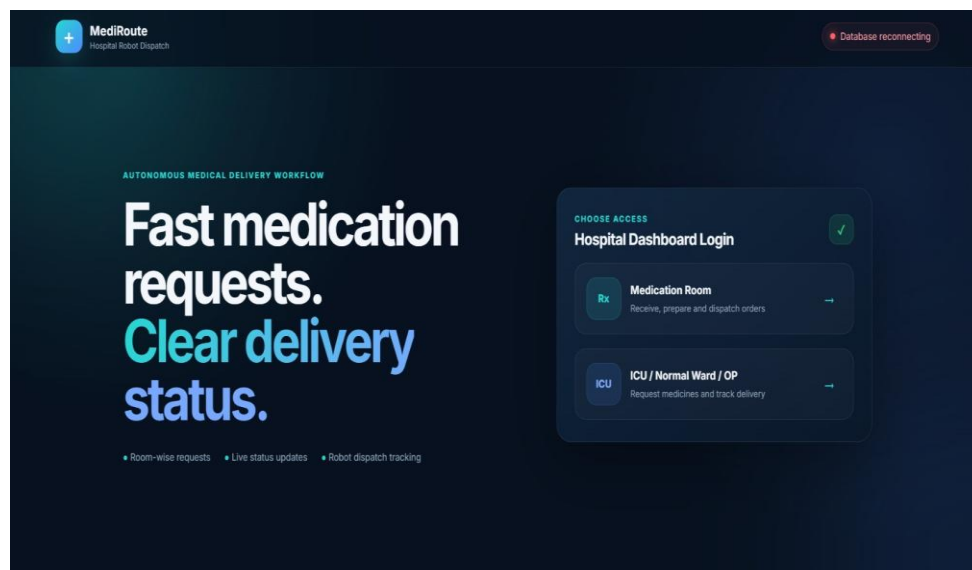


Figure 6.9 Dispatch Status and Monitoring

CHAPTER 7

PROTOTYPE DEVELOPMENT

7.1 CONCEPT AND DESIGN

The prototype was conceived as a compact, two-wheel differential-drive robot capable of carrying a small medical payload while autonomously following a marked corridor path and stopping safely for obstacles. The initial design phase involved sketching the chassis layout, deciding sensor placement for reliable line detection and forward-facing obstacle detection, and selecting components based on cost, availability, and compatibility with the Arduino UNO platform.

7.2 CHASSIS FABRICATION

The chassis was fabricated using lightweight cardboard reinforced with acrylic sheet, cut and assembled into a two-tier structure. The lower tier was designed to mount the drive motors, wheels, and battery pack, while the upper tier was designed to hold the control electronics and the medical storage compartment. Mounting points for the IR sensors and ultrasonic sensor were cut into the front section of the lower and upper tiers respectively, ensuring clear, unobstructed sensing.

7.3 COMPONENT ASSEMBLY

Following chassis fabrication, the Arduino UNO, L298N motor driver, voltage regulator, and battery pack were mounted and secured using adhesive and fasteners. All modules were wired according to the circuit diagram presented in Appendix 1, with jumper wires routed neatly along the chassis to avoid interference with the wheels and moving parts.

7.4 SENSOR AND MOTOR INSTALLATION

The two IR sensors were installed on the underside of the front chassis section, positioned symmetrically about the centerline and at a height calibrated to reliably detect the tape path. The ultrasonic sensor was mounted at the front edge of the upper chassis tier, facing forward and level with the ground plane, to ensure consistent obstacle detection along the robot's direction of travel. The two DC gear motors were installed on the lower chassis tier, coupled to the rear wheels, with a caster or free-rolling wheel arrangement at the front for stability.

7.5 FINAL PROTOTYPE

The completed prototype after full assembly, positioned on the simulated hospital corridor test track. The front-mounted IR sensor pair and ultrasonic sensor are visible at the lower front of the chassis, while the LCD display, push button, and control electronics are mounted on the upper deck for easy visibility and access during testing.

7.6 3D CAD MODELING

7.6.1 Isometric view

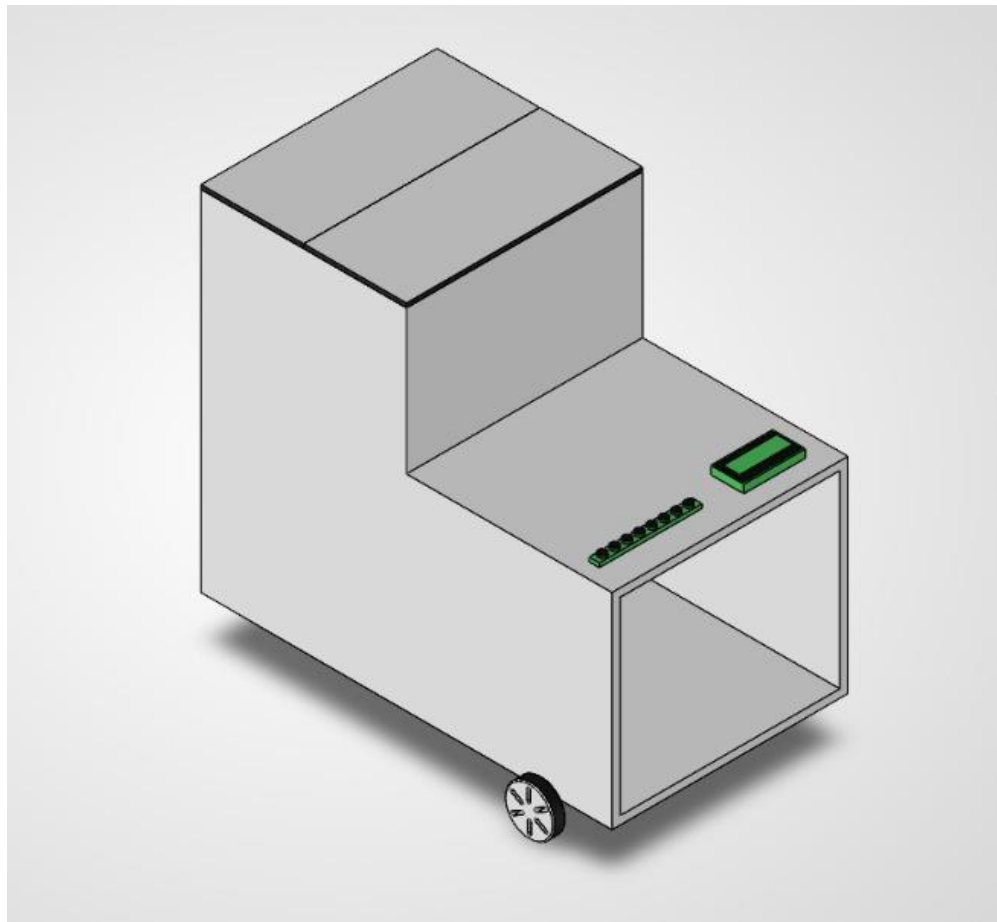


Figure7.6.1 Isometric View

7.6.2 Left side view

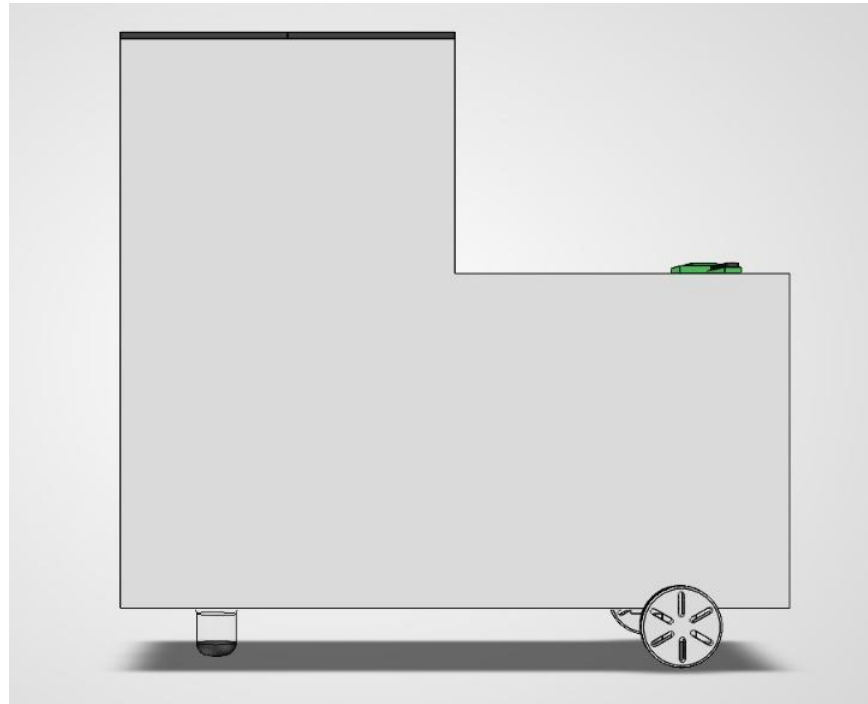


Figure 7.6.2 Left Side View

7.6.3 Top view

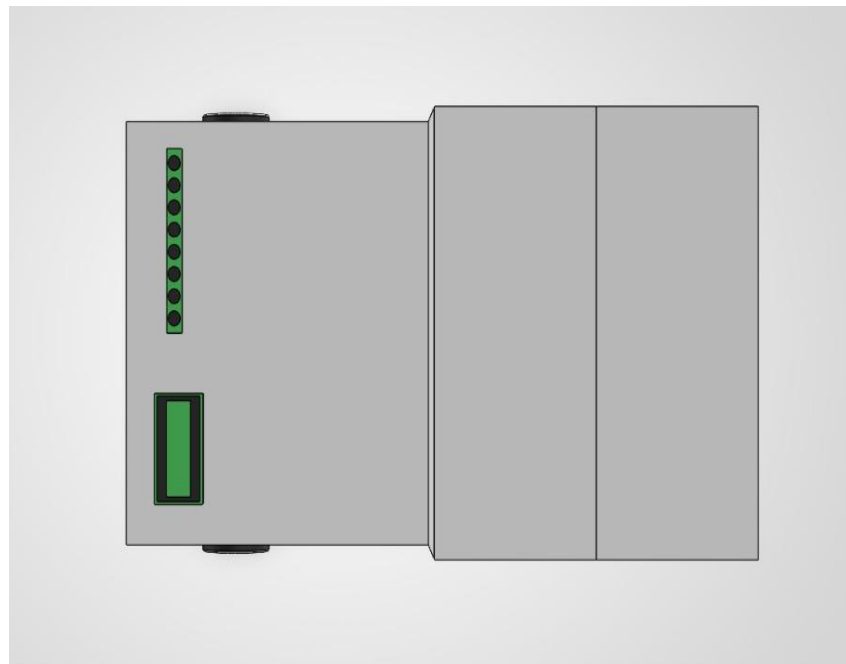


Figure 7.6.3 Top View

7.6.4 Front view

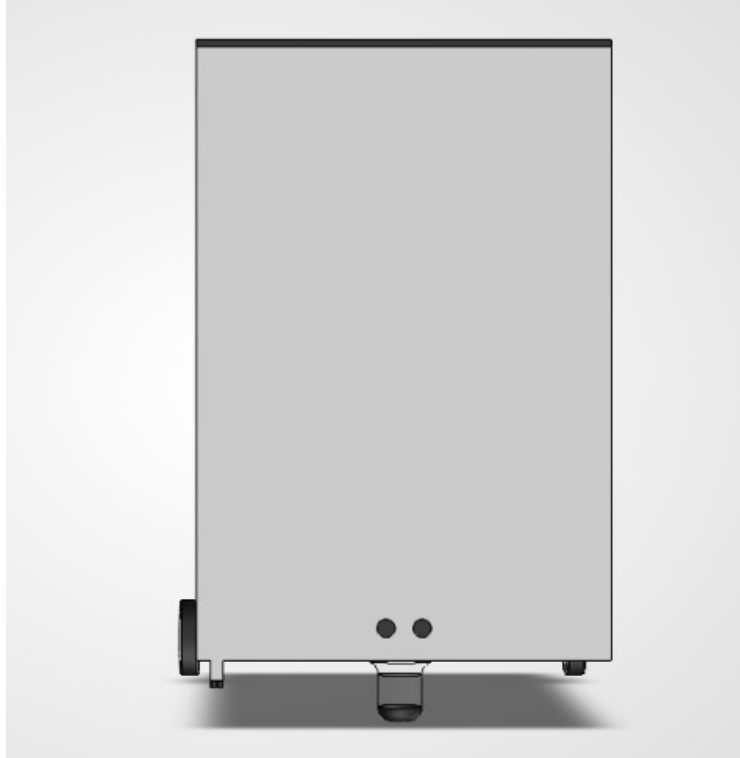


Figure 7.6.4 Front View

7.7 FINAL PROTOTYPE VIEWS

7.7.1 Front view

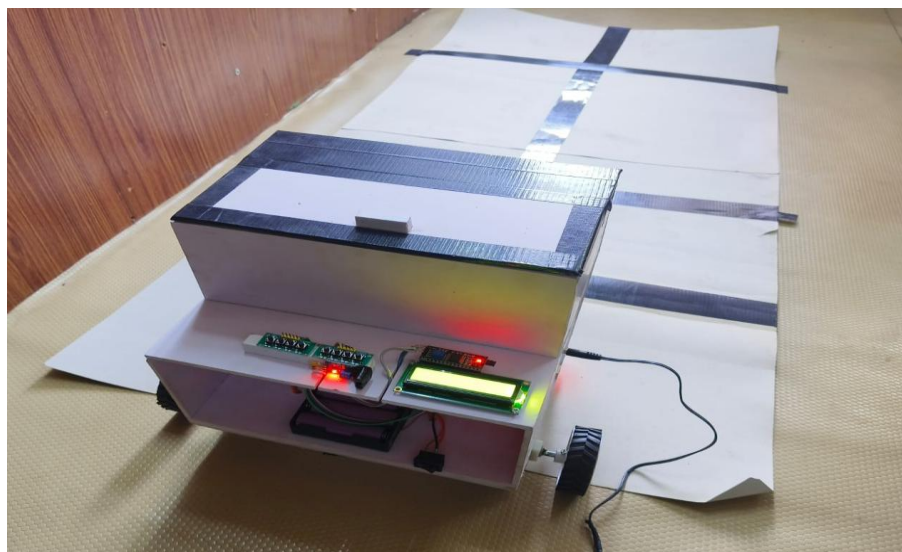


Figure 7.7.1 Front View

7.7.2 Top view



Figure 7.7.2 Top View

7.7.3 Isometric View



Figure 7.7.3 Isometric View

CHAPTER 8

RESULTS AND DISCUSSION

The completed prototype was subjected to a series of controlled experiments on the simulated hospital corridor test track to evaluate its obstacle detection, navigation accuracy, turning and stopping behavior, payload-carrying capability, and battery performance. Each test was repeated multiple times to check for consistency, and the results are summarized in this chapter.

8.1 OBSTACLE DETECTION TEST

An obstacle was placed at varying distances directly in the robot's path to evaluate the reliability of the ultrasonic-sensor-based stopping behavior. In every trial, the robot detected the obstacle and came to a complete stop before contact, with the stopping distance closely matching the 15 cm safety threshold programmed into the firmware.

Table 8.1 Obstacle Detection Test Results

Trial No.	Obstacle Placed At (cm)	Robot Stopped At (cm)	Result
1	30	16.2	Pass
2	25	15.8	Pass
3	20	14.9	Pass
4	35	16.5	Pass
5	15	14.5	Pass

8.2 NAVIGATION TEST

Eight repeated trial runs were conducted over a fixed 3-metre section of the corridor path (with no obstacles present) to evaluate line-following accuracy. The robot completed the route in approximately 14 seconds on average, maintaining an average lateral deviation from the centre of the marked path of under 2.2 centimeters in every run.

Table 8.2 Navigation Test - Trial Run Data

Trial No.	Distance (m)	Time Taken (s)	Avg. Deviation (cm)
1	3.0	14.2	1.8
2	3.0	13.9	1.6
3	3.0	14.5	2.1
4	3.0	14.1	1.7
5	3.0	13.8	1.5
6	3.0	14.3	1.9
7	3.0	14.6	2.2
8	3.0	14.0	1.6

8.3 TURNING AND STOPPING TEST

The robot's ability to negotiate the perpendicular junction on the test track and to stop precisely at the end of a delivery route was evaluated over five trials. In all trials, the robot continued straight through the junction along the intended primary path and stopped within a small margin of the marked endpoint.

Table 8.3 Turning and Stopping Test - trial run data

Trial No.	Junction Negotiated Correctly	Stopping Overshoot (cm)
1	Yes	2.1
2	Yes	1.8
3	Yes	2.4
4	Yes	1.6

8.4 PAYLOAD TEST

The medical storage compartment was loaded with increasing payload weights to determine the maximum load the prototype could carry while maintaining stable navigation. Performance remained stable up to approximately 400 grams, beyond which increased wheel slip and reduced motor speed were observed.

Table 8.4 Payload test results

Payload Weight (g)	Navigation Stability	Observation
100	Stable	No noticeable change in speed or path tracking
200	Stable	No noticeable change in speed or path tracking
300	Stable	Slight reduction in acceleration
400	Stable	Noticeable but acceptable reduction in speed
500	Unstable	Wheel slip observed on turns, path deviation increased

8.5 BATTERY PERFORMANCE

The battery voltage was monitored over a continuous 40-minute operating period to assess power stability and estimate usable run time. The voltage declined gradually and predictably, with the voltage regulator maintaining a stable 5V supply to the control electronics throughout the test despite the falling battery voltage.

Table 8.5 Battery performance data

Time (min)	Battery Voltage (V)
0	7.45
10	7.36
20	7.28
30	7.19

8.6 OVERALL PERFORMANCE

Taken together, the results confirm that the prototype reliably detects and stops for obstacles, tracks the marked corridor path with minimal deviation, negotiates junctions and stops accurately at delivery endpoints, carries a modest medical supply payload without instability, and maintains stable control electronics operation over a practical working duration on a single charge. These results validate the feasibility of the proposed low-cost, sensor-guided approach for autonomous medical supply delivery within a fixed hospital corridor layout.

CHAPTER 9

SAFETY AND MAINTENANCE

9.1 SAFETY FEATURES

The prototype incorporates several basic safety features to ensure predictable and safe operation in a shared corridor environment. The ultrasonic-sensor-based obstacle detection system immediately halts both drive motors whenever an object is detected within the preset safety distance, preventing collisions with people or equipment. The push-button start/stop control ensures the robot only begins moving when deliberately activated by a staff member, and the LCD status display provides continuous visual confirmation of the robot's operating state, reducing the chance of unexpected movement going unnoticed.

9.2 BATTERY AND ELECTRICAL SAFETY

The Li-ion battery pack is connected through the voltage regulator to protect the control electronics from over-voltage conditions, and all wiring is routed and secured to avoid short circuits or accidental disconnection during movement. As with any lithium-ion battery system, the pack should be charged using a charger matched to its specification, should not be left charging unattended for extended periods, and should be inspected periodically for swelling, damage, or overheating, with damaged cells replaced immediately rather than continued in use.

9.3 MAINTENANCE PROCEDURE

- Inspect wheels, motor mounts, and chassis fasteners periodically for looseness caused by vibration during operation.
- Clean the IR sensor and ultrasonic sensor lenses regularly to prevent dust accumulation from affecting detection accuracy.
- Check and recalibrate IR sensor threshold values if the test track surface or lighting conditions change significantly.
- Monitor battery voltage before each operating session and recharge before it falls below the manufacturer's recommended minimum.
- Verify all wiring connections periodically, particularly at the motor driver terminals, which experience the highest current draw.
- Update and re-upload firmware as needed when control parameters (such as the proportional gain K_p) require adjustment.

9.4 ADVANTAGES:

- ✓ Reduces manual effort involved in transporting medical supplies within hospitals.
- ✓ Provides autonomous movement for indoor medical supply delivery.
- ✓ Reduces the workload of nurses and hospital support staff.
- ✓ Detects obstacles and helps avoid collisions during navigation.
- ✓ Minimizes unnecessary human movement inside hospital corridors.
- ✓ Can be further upgraded with advanced navigation and IoT technologies.
- ✓ Suitable for transporting medicines and other lightweight medical supplies.

9.5 APPLICATIONS:

- ✓ Medicine delivery from pharmacy to hospital wards.
- ✓ Delivery of laboratory samples within hospitals.
- ✓ Smart hospital and healthcare automation systems.
- ✓ Isolation wards and controlled-access hospital areas.
- ✓ Research and demonstration of autonomous mobile robots.
- ✓ Transportation of medical supplies between different departments.
- ✓ Transportation of documents and essential hospital materials.
- ✓ Laboratories and diagnostic departments.

CHAPTER 10

APPENDICES

APPENDIX – 1

10.1 CIRCUIT DIAGRAM

Figure A1.1 shows the complete wiring diagram of the prototype, mapping each sensor, display, and motor driver connection to its corresponding Arduino UNO pin, as implemented in the fabricated prototype and referenced throughout Chapter 4.

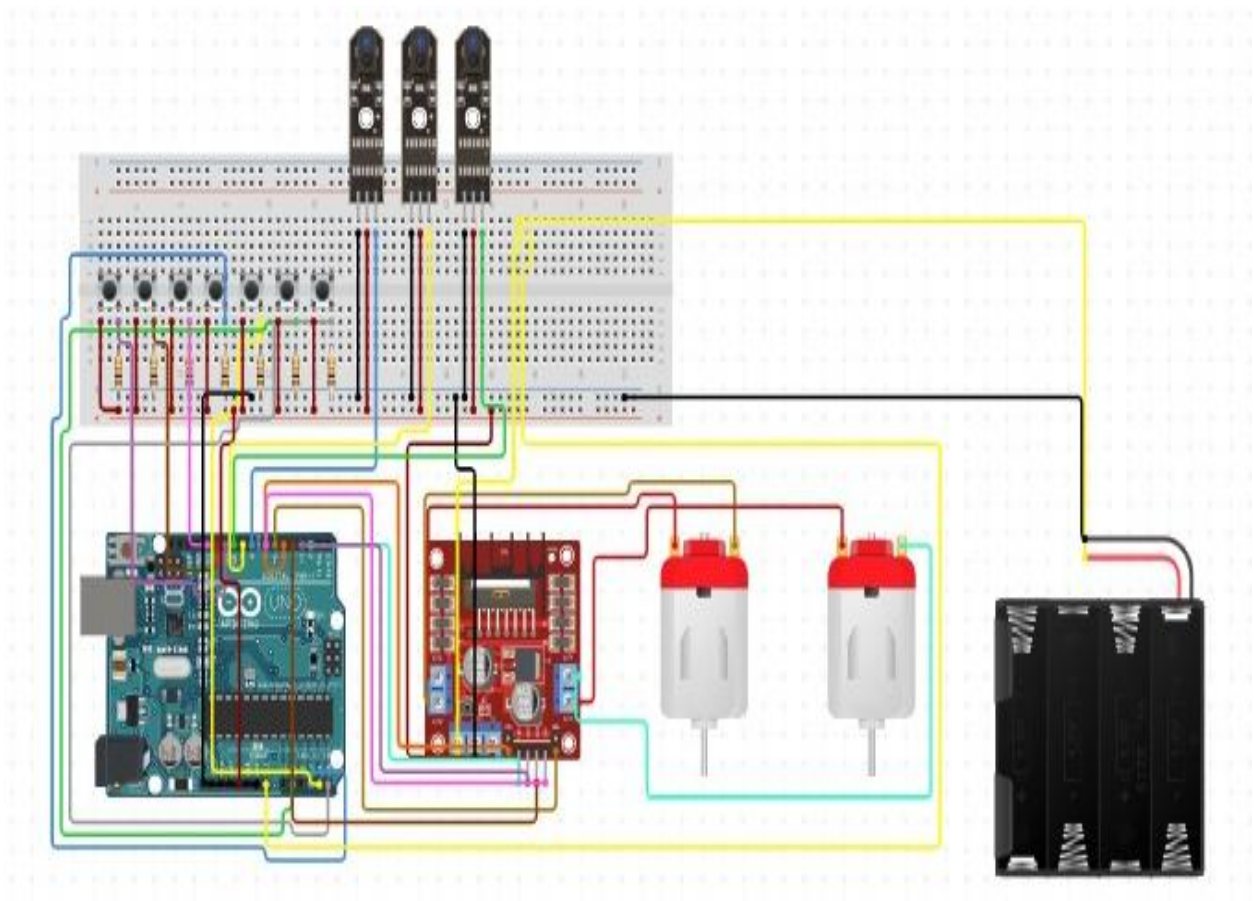


Figure 10.1 Circuit / Wiring Diagram

APPENDIX – 2

COMPONENTS SPECIFICATION

PROTOTYPE SPECIFICATION

1. Arduino UNO

Microcontroller: ATmega328P

Operating Voltage: 5 V

Digital I/O Pins: 14

Analog Input Pins: 6

Application: Main controller for zone selection, sensor monitoring, navigation, and motor control

2. IR Sensor – Path Detection

Sensor Type: Infrared sensor

Operating Voltage: 3.3–5 V

Output: Digital

Application: Detecting path columns and assisting robot navigation

3. IR Sensor – Obstacle Detection

Sensor Type: Infrared obstacle detection sensor

Operating Voltage: 3.3–5 V

Output: Digital

Application: Detecting obstacles and stopping the robot automatically

4. DC Motor

Type: DC geared motor

Operating Voltage: 12 V

Function: Provides movement to the robot

Application: Forward, reverse, left, and right movement

5. Motor Driver Module

Type: Dual-channel motor driver

Operating Voltage: As per motor driver specification

Function: Controls the direction and speed of DC motors

Application: Interfacing the Arduino UNO with the DC motors

6. I2C LCD Display

Display Type: 16 × 2 LCD

Interface: I2C

Operating Voltage: 5 V

Application: Displaying zone selection, path status, obstacle alerts, and delivery status

7. Zone Selection Switches

Type: Push button switches

Number of Inputs: 7

Function: Selection of Zones 1–6 and Home position

Application: Providing the destination input to the robot

8. Battery

Type: Rechargeable battery

Voltage: As per motor and circuit requirements

Function: Provides electrical power to the robot

Application: Power supply for the motors, Arduino, sensors, and other electronic components

9. Robot Chassis

Material: Lightweight structural material

Properties: Lightweight, rigid, and easy to fabricate

Application: Supporting the electronic components, motors, wheels, and medical supply compartment

10. Wheels

Type: Rubber wheels

Properties: Good traction and smooth movement

Application: Enables movement of the robot along the predefined path

11. Medical Supply Storage Compartment

Material: Lightweight enclosed material

Properties: Compact, lightweight, and suitable for carrying medical supplies

Application: Carrying medicines and other required medical supplies to the selected hospital zone

APPENDIX – 3

PSEUDO CODE

The following pseudocode describes the main control logic of the **Autonomous Medical Supply Delivery Robot for Hospitals**. The program manages zone selection, path detection, autonomous navigation, obstacle avoidance, motor control, LCD status display, and return-to-home operation.

The program controls the overall operation of the robot, including zone selection, path navigation, IR sensor monitoring, obstacle detection, DC motor control, LCD status display, and return-to-home operation.

The Arduino program is structured using the standard `setup()` and `loop()` functions. The `setup()` function initializes the input and output pins, I2C LCD display, serial communication, sensors, and motor control pins. The `loop()` function continuously monitors the zone selection inputs and controls the robot according to the selected destination.

The program uses separate functions such as `Zone Path()`, `Obstacle()`, `MotorForwardCmd()`, `MotorReverseCmd()`, `MotorLeftCmd()`, `MotorRightCmd()`, and `MotorStopCmd()` to perform navigation, obstacle detection, and motor movement operations. The `LCD_I2C.h` library is used to interface the 16×2 I2C LCD display with the Arduino.

START

1. SYSTEM INITIALIZATION

Power ON the robot

Initialize Arduino controller

Initialize I2C LCD display

Initialize Zone 1–6 selection inputs

Initialize Home selection input

Initialize path-detection IR sensor

Initialize obstacle-detection IR sensor

Initialize left and right DC motors

Set motor speed

Display project title on LCD

Display "Initialized Successfully"

2. WAIT FOR DESTINATION

Continuously read all zone selection inputs

Check whether Zone 1, Zone 2, Zone 3,
Zone 4, Zone 5, Zone 6, or Home is selected

3. ZONE SELECTION

IF Zone 1 is selected:

Set Zone = 1

Record selected zone

Reset path counter

Display "ZONE 1 Selected"

Move forward

ELSE IF Zone 2 is selected:

Set Zone = 2

Record selected zone

Reset path counter

Display "ZONE 2 Selected"

Move forward

ELSE IF Zone 3–6 is selected:

Set the corresponding zone number

Record selected zone

Reset path counter

Display selected zone

Move forward

ELSE IF Home is selected:

Set Home navigation mode

Display "Home Selected"

Start return movement

4. PATH DETECTION

Read the path IR sensor continuously

Detect each path column

Increment the path counter

Display:

"Path Column = [Count]"

"Zone = [Zone Number]"

5. DESTINATION NAVIGATION

IF required path column is reached:

Continue forward for the specified distance

Perform required turning operation

Turn RIGHT for designated zones

Turn LEFT for designated zones

Continue moving forward

Check obstacle sensor during movement

6. OBSTACLE DETECTION

Read obstacle IR sensor

IF obstacle is detected:

Immediately stop the motors
Display "OBSTACLE FOUND"
Display "ROBOT STOPPED"
Wait until obstacle is removed
Display "OBSTACLE CLEAR"
Resume navigation

ELSE:

Continue normal movement

7. ZONE ARRIVAL

Complete the required navigation sequence
Stop the motors
Display "ZONE REACHED"
Keep the robot stationary
Wait for the next command

8. RETURN TO HOME

IF Home is selected:

Start reverse movement
Follow the return path
Use the recorded destination zone
to determine the turning direction
Turn LEFT or RIGHT as required
Continue reverse movement
Monitor obstacle sensor continuously

9. HOME ARRIVAL

Detect the recorded path position

Stop the motors

Display "ZONE REACHED"

Clear the LCD

Reset Zone and navigation variables

10. MOTOR CONTROL

Forward:

Drive both motors forward

Reverse:

Drive both motors in reverse

Left Turn:

Control left and right motors
in opposite directions

Right Turn:

Control left and right motors
in opposite directions

Stop:

Stop both motors immediately

11. SPEED DISPLAY

Convert motor speed value into percentage

Display speed percentage on LCD when required

12. REPEAT OPERATION

Return to destination selection

Wait for the next delivery request

Repeat the complete process

END

CHAPTER 11

CONCLUSION

This project successfully demonstrated the design, fabrication, and experimental validation of a low-cost autonomous medical supply delivery robot for hospital corridor logistics. The prototype combined IR-sensor-based line-following navigation with ultrasonic obstacle detection, controlled by an Arduino UNO, to achieve reliable, contact-free point-to-point delivery of a medical supply payload along a fixed corridor route. Across all experimental tests - obstacle detection, navigation accuracy, turning and stopping, payload carrying, and battery performance - the prototype demonstrated consistent, repeatable behavior, confirming that a simple, inexpensive combination of IR and ultrasonic sensing is sufficient to meet the core safety and reliability requirements of a fixed-route hospital delivery task. The total fabrication cost of approximately Rs. 2,820 further confirms that this approach is significantly more affordable than existing commercial hospital delivery robots, making it particularly relevant for small and medium-sized healthcare facilities with limited budgets. The safety and maintenance considerations outlined in Chapter 8, together with the future scope identified in Chapter 9 - including IoT-based monitoring, RFID ward identification, advanced SLAM-based navigation, and full hospital system integration - provide a clear roadmap for evolving this validated prototype into a deployable clinical solution. Overall, this project confirms that embedded-systems-based autonomous delivery, incorporating basic obstacle avoidance, is a practical, scalable, and cost-effective direction for reducing manual workload and improving safety in hospital material handling.

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